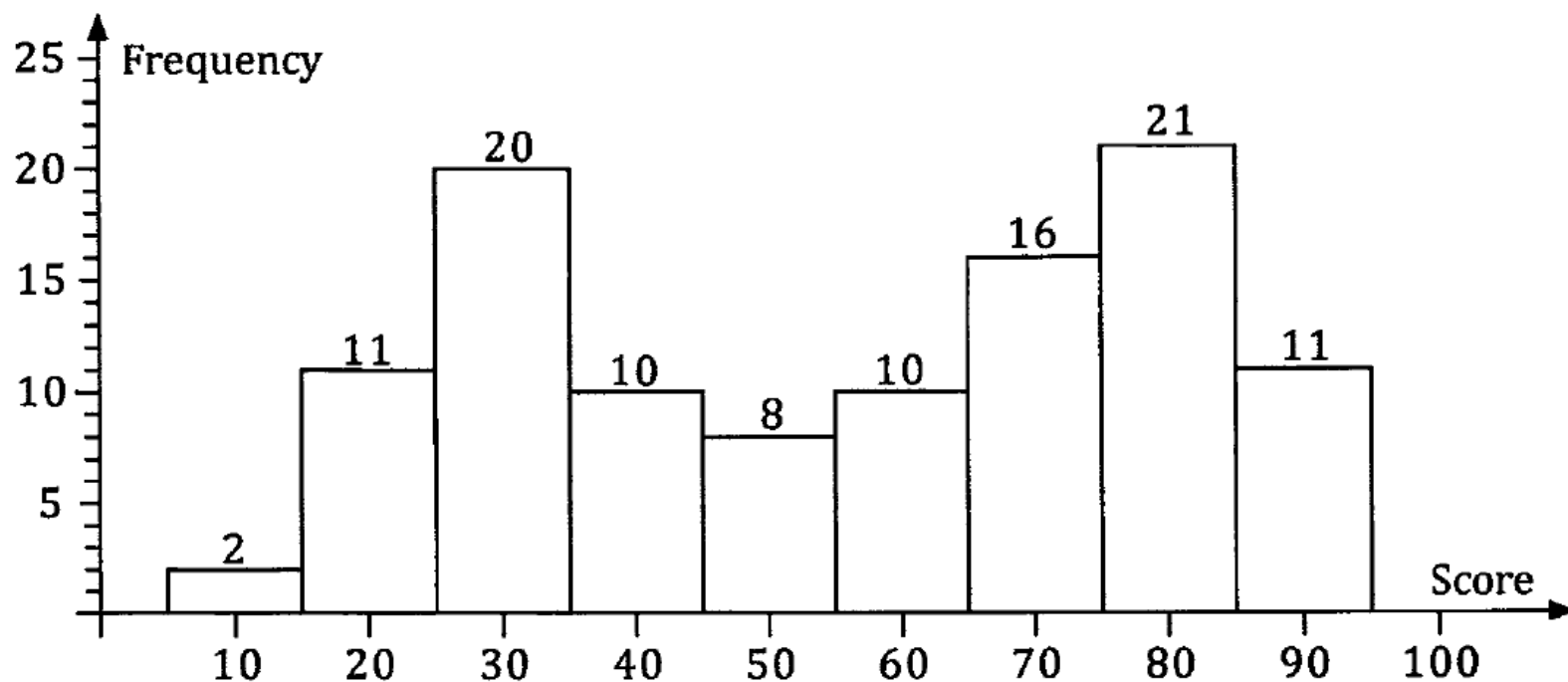


Describing a distribution  
of a score

The histogram below shows the distribution of scores achieved by the students of a school in a mathematics exam.



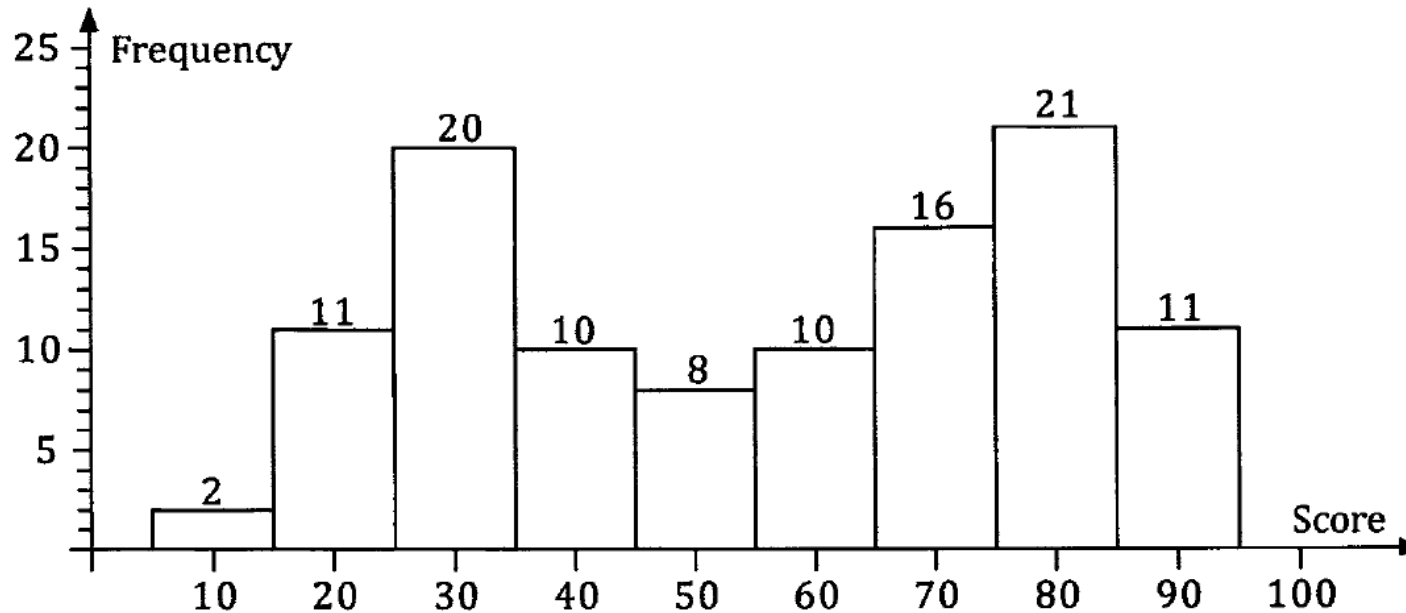
# Description

- Location – mean and median
- Spread – Range of scores gives information about their spread
- Shape – symmetry, gaps, more/less dense regions, outliers, modality, convey information about the shape of a distribution
- Anything else of relevance

# Use words such as

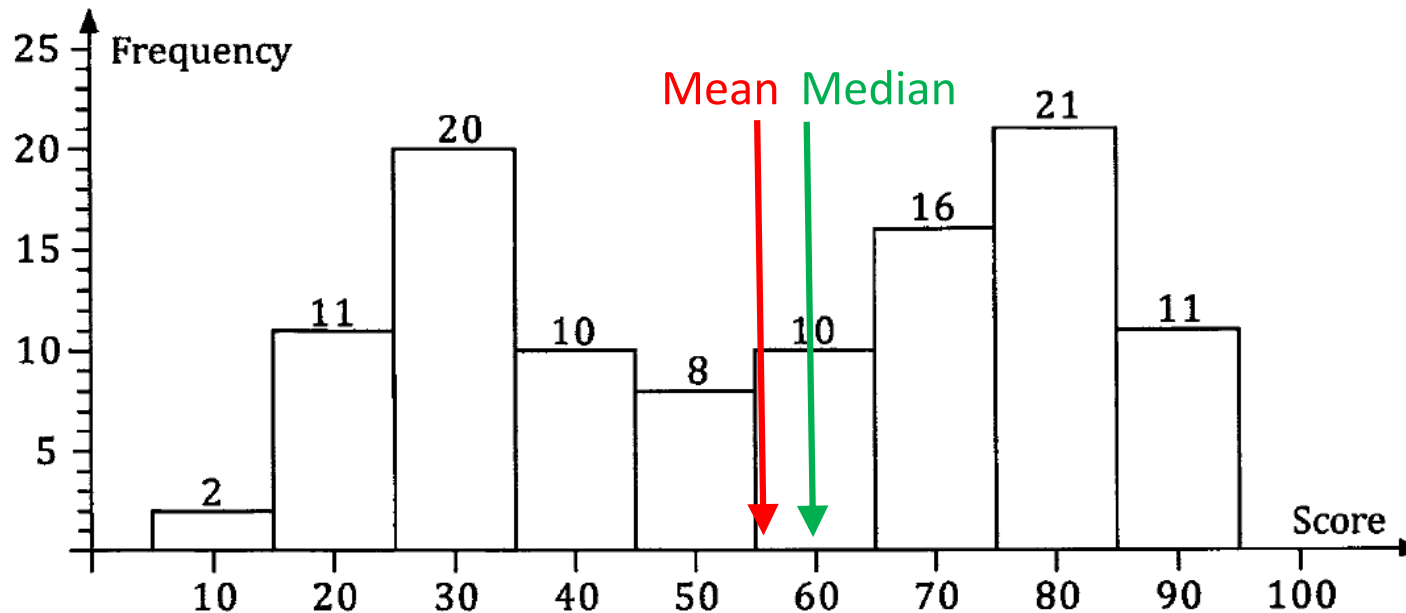
- Lowest score
- Highest score
- Gaps
- Tightly packed
- Spread out
- Clusters
- Groups
- Symmetrical
- Outliers
- Uniform

# Relevant information



One hundred and nine (109) students from the school sat the examination.

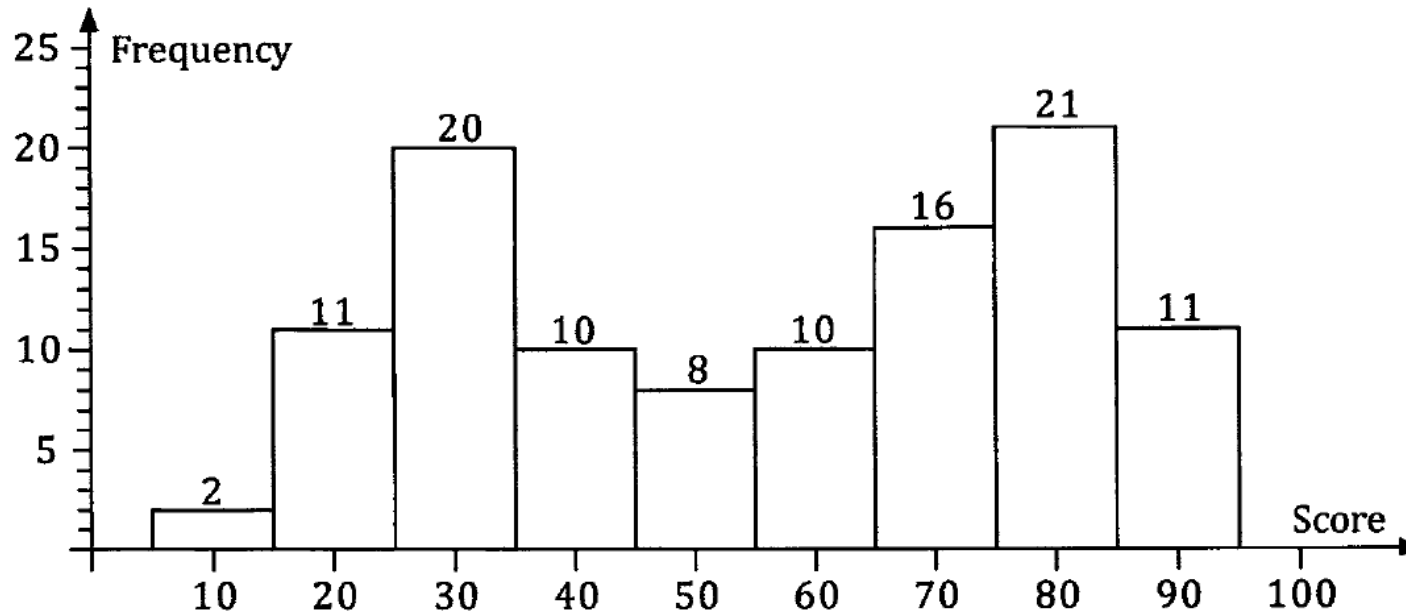
# Location



$$\text{Mean} = \frac{2 \times 10 + 11 \times 20 + \dots + 11 \times 90}{109} = 55.3$$

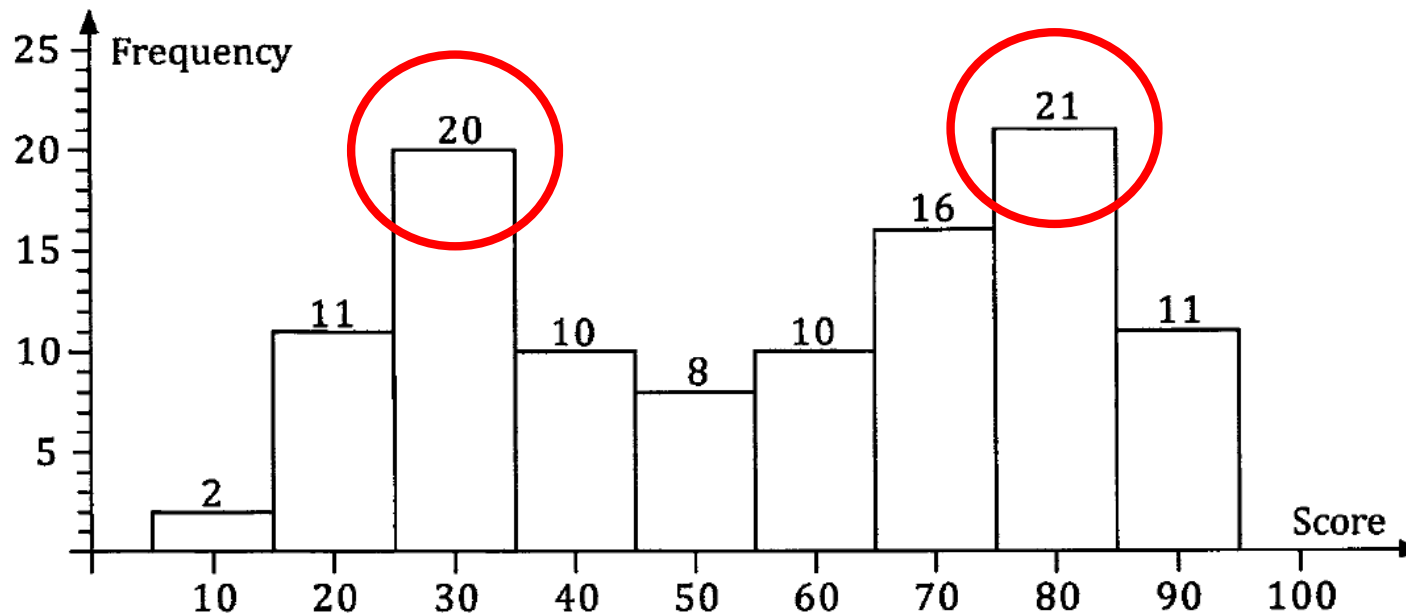
Median – 55<sup>th</sup> student, scored between 55 and 65

# Spread



Range:  $95 - 5 = 90$

# Shape



Spread reasonably symmetrical

Bi-modal, peak around 30 and again at 80



One hundred and nine students from the school sat the exam.

← Any relevant information perhaps not covered in location, spread and shape?

Using the mid-point of each class interval gives an estimated mean of 55.3 and the middle ranked student, i.e. the 55th student, achieved a mark between 55 and 65.

← Comment about location.

The scores were well spread out from about 5 to 95, i.e. the range was about 90.

← Comment about spread.

The scores were reasonably symmetrically spread about a mid point of about 55. The distribution of scores was bimodal peaking around 30 and again around 80.

← Comment on shape.

Approximately 30% of the students scored less than 35 and approximately 30% scored more than 75.

← Anything else you notice of relevance.

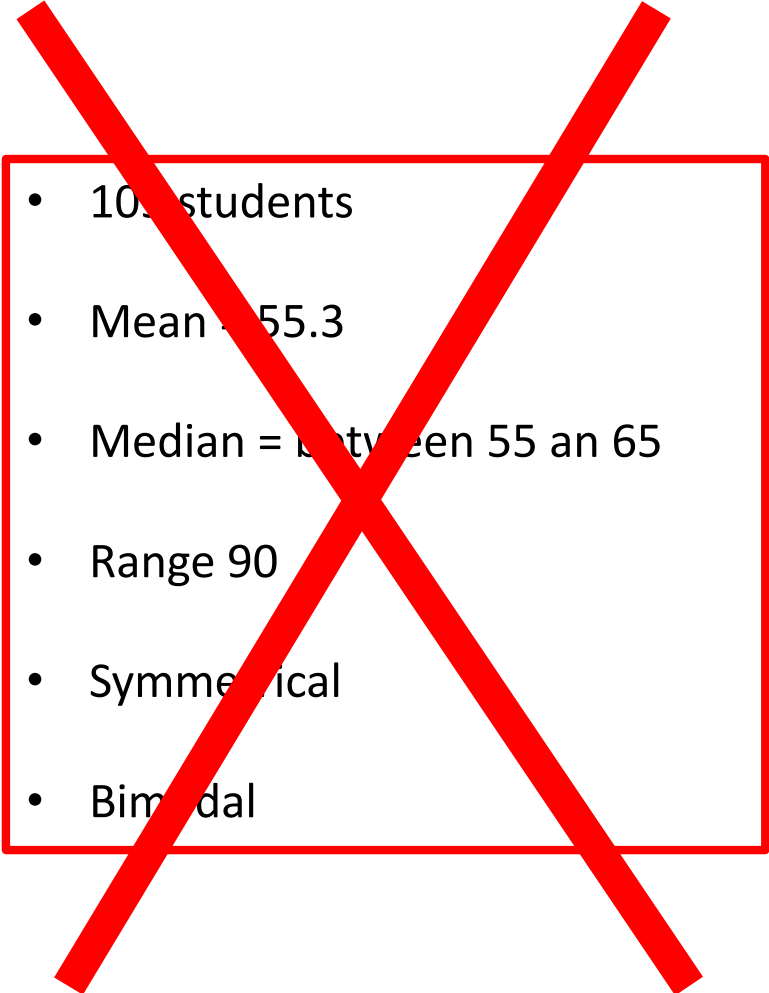
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The scores were reasonably symmetrically spread about a mid point of about 55. The distribution of scores was bimodal peaking around 30 and again around 80.

Approximately 30% of the students scored less than 35 and approximately 30% scored more than 75.

- 
- 109 students
  - Mean = 55.3
  - Median = between 55 and 65
  - Range 90
  - Symmetrical
  - Bimodal